Verilock

A Password-Protected Word Display System

Lab Report

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1. Introduction

Field Programmable Gate Arrays (FPGAs) have become essential tools in modern digital system design, offering unparalleled flexibility in implementing custom hardware solutions. This report presents the design and implementation of Verilock, a password-protected word display system developed on the Digilent Basys3 FPGA development board. The project demonstrates the practical application of finite state machine (FSM) design, sequential logic circuits, and human-machine interface principles.

The Verilock system addresses a common requirement in embedded systems: securing user data while providing an intuitive interface for interaction. By combining directional pattern-based authentication with a comprehensive text editing capability, the project showcases how relatively simple digital components can be orchestrated to create a functional and secure system.

The motivation behind this project stems from the need to understand fundamental digital design concepts through hands-on implementation. Rather than relying solely on theoretical knowledge, building a complete system from specification to hardware deployment provides invaluable experience in dealing with real-world constraints such as timing, resource utilization, and user interface design.

2. Project Objectives

The primary objectives of the Verilock project are defined as follows:

1. To design and implement a secure authentication mechanism using directional button patterns as passwords, demonstrating pattern recognition and verification in hardware.
2. To develop a text editing interface that allows users to create, modify, and navigate through custom messages displayed on 7-segment displays.
3. To implement a finite state machine architecture that manages system states, transitions, and user interactions in a predictable and reliable manner.
4. To demonstrate proficiency in Verilog HDL coding for synthesizable digital designs targeting FPGA platforms.
5. To integrate multiple hardware components including buttons, switches, LEDs, and 7-segment displays into a cohesive user interface.
6. To provide dual operational modes (looping and infinite) for message navigation, enhancing user flexibility and demonstrating configurable system behavior.

3. System Overview

3.1 System Architecture

The Verilock system architecture follows a modular design philosophy, organizing functionality into distinct logical units that communicate through well-defined interfaces. The top-level module integrates all subsystems and manages the overall data flow within the design.

The system operates on a single 100 MHz clock domain, with all sequential elements synchronized to this master clock. This simplifies timing analysis and eliminates the need for clock domain crossing logic, which would add complexity and potential sources of errors.

[Insert System Architecture Diagram Here]

Figure 1: System Architecture Block Diagram

3.2 Main Components

The Verilock system comprises six primary functional modules, each responsible for a specific aspect of system operation:

Module	Function
Input Handler Unit	Processes D-pad button inputs and generates edge-detected signals for state machine transitions
Password Manager	Stores the 6-direction pattern password and performs verification against user input
Menu Control Unit	Controls system states and manages transitions between SETP and WORD modes
Character Database	Stores 28 character definitions (A-Z, space, null) with corresponding 7-segment encodings
Display Controller	Manages 7-segment display multiplexing and character rendering with cursor blinking
Word Editor	Manages character selection, cursor navigation, and message editing with viewport control

Table 1: Verilock System Module Description

4. Hardware Specifications

4.1 Development Platform

The Verilock system is implemented on the Digilent Basys3 development board, which features the Xilinx Artix-7 FPGA (XC7A35T-1CPG236C). This platform provides an ideal balance of resources and cost for educational and prototyping purposes. The board includes all necessary peripherals for this project without requiring external components.

FPGA Specifications:

- Logic Cells: 33,280 (sufficient for complex state machines and memory structures)
- Block RAM: 1,800 Kbits (utilized for message storage)
- Clock Management Tiles: 5 (enabling flexible clock generation)
- System Clock: 100 MHz oscillator (W5 pin)

4.2 On-Board Components Utilized

The following on-board components are employed in the Verilock implementation:

D-Pad Buttons (5 buttons):

The five push-buttons arranged in a directional pad configuration serve as the primary input interface. Each button is active-high and directly connected to FPGA I/O pins.

- BTNU (Up) - Pin T18: Password entry, character increment, menu navigation
- BTND (Down) - Pin U17: Password entry, character decrement, menu navigation
- BTNL (Left) - Pin W19: Password entry, cursor movement left
- BTNR (Right) - Pin T17: Password entry, cursor movement right
- BTNC (Center) - Pin U18: Confirmation, menu selection, exit to menu

Toggle Switch:

- SW0 (Pin V17): Mode selection switch - 0 for looping mode (skips null characters), 1 for infinite mode (traverses all 512 positions)

7-Segment Displays:

Four multiplexed 7-segment displays with common anodes provide visual output. The displays use active-low segment cathodes (seg[6:0]) and active-low anode control (an[3:0]). A decimal point output (dp) is also available for additional indicators.

LEDs:

Sixteen user LEDs (led[15:0]) are available for status indication. In the current implementation, LED0 displays the state of SW0 to indicate the active navigation mode.

5. Design Methodology

5.1 Development Approach

The development of Verilock followed a structured top-down design methodology. The process began with comprehensive specification of system requirements, followed by architectural planning, module design, implementation, and finally hardware validation. This approach ensured that each component was designed with a clear understanding of its role within the overall system.

5.2 State Machine Design

The core of the Verilock system is a finite state machine implemented using a single always block sensitive to the positive edge of the system clock. Eight distinct states govern system operation:

State	Encoding	Description
STATE_MENU	3'b000	Main menu display
STATE_SETP_AUTH	3'b001	Password authentication for SETP
STATE_SETP_NEW	3'b010	New password entry
STATE_SETP_DONE	3'b011	Password change confirmation
STATE_WORD_AUTH	3'b100	Password authentication for WORD
STATE_WORD_EDIT	3'b101	Message editing mode
STATE_ERROR	3'b110	Error display state
STATE_SEARCHING	3'b111	Non-null character search

Table 2: FSM State Definitions

5.3 Password Encoding Scheme

The password system employs a 2-bit encoding for each of the four directional inputs. A complete password consists of six direction inputs, requiring 12 bits of storage. The encoding scheme is as follows: UP = 2'b00, DOWN = 2'b01, LEFT = 2'b10, RIGHT = 2'b11. The default password is set to six consecutive UP presses (12'b00_00_00_00_00_00).

5.4 Display Multiplexing

The four 7-segment displays share common segment lines and are controlled through time-division multiplexing. A 17-bit counter generates a refresh tick approximately every 650 microseconds (65,000 clock cycles at 100 MHz). Each display is active for one refresh period

in sequence, creating a refresh rate of approximately 385 Hz per digit. This rate is sufficiently high to eliminate visible flicker while remaining within the display's response time.

5.5 Character Encoding

Characters are encoded using 5-bit values, allowing representation of 32 unique symbols. The character set includes the 26 English alphabet letters (0-25), a space character (26), and a null/blank character (27). Each character maps to a unique 7-segment pattern through a combinational lookup function. The segment outputs use active-low logic, where a '0' illuminates a segment.

6. Implementation Details

6.1 Input Processing

Button inputs are processed using simple edge detection without hardware debouncing. Each button signal is registered on every clock edge, and a rising edge is detected by comparing the current value with the previous cycle's value. This simplified approach works effectively for the Basys3 board's integrated button conditioning circuits. The edge detection logic generates single-cycle pulse signals (BTNU_pressed, BTND_pressed, etc.) that trigger state transitions and actions.

6.2 Message Storage

The system maintains a 512-character message buffer, with each character stored as a 5-bit value. This allows for substantial message capacity while utilizing FPGA block RAM efficiently. The message array is initialized to null characters at system startup. A 9-bit cursor position register tracks the current editing location, while a separate 9-bit viewport_start register manages which four characters are currently displayed.

6.3 Viewport Management

The four-digit display presents a sliding window view of the message buffer. The viewport automatically adjusts to keep the cursor visible: if the cursor moves left of the viewport, viewport_start shifts left; if the cursor moves beyond the third position, viewport_start advances to keep the cursor at the rightmost position. This creates a natural scrolling behavior as the user navigates through longer messages.

6.4 Navigation Modes

Two navigation modes provide flexibility in cursor movement:

Looping Mode (SW0 = 0): The cursor skips null characters, jumping directly to the next or previous non-null character. This mode enters the STATE_SEARCHING state which iterates through the message buffer until a valid character is found or 512 positions have been checked.

Infinite Mode (SW0 = 1): The cursor moves to every position regardless of character content, wrapping from position 511 to 0 or vice versa. This mode is useful for inserting characters at specific positions in a sparse message.

6.5 Cursor Indication

The cursor position is indicated through a blinking effect on the current character. A 25-bit counter generates a blink signal at approximately 1.5 Hz (blink_counter[24]). During the edit

state, the digit corresponding to the cursor position alternates between displaying its character and being blank, providing clear visual feedback of the editing position.

6.6 Timing and Temporary States

Temporary states (STATE_SETP_DONE and STATE_ERROR) use a 27-bit timer initialized to 100,000,000 cycles, providing a one-second display duration at 100 MHz. The timer decrements each clock cycle, and when it reaches zero, the system automatically transitions back to the menu state. This provides visual confirmation of successful operations or error conditions.

7. Project Flowchart

The system flow diagram illustrates the complete operational sequence of the Verilock system. Starting from the main menu, users can select between two primary modes: SETP for password management and WORD for message editing. Both modes require successful password authentication before granting access to their respective functions.

[Insert Project Flowchart Here]

Figure 2: Verilock System Flow Diagram

The flowchart demonstrates the state transitions and decision points within the system. Key features include the password verification branches, error handling paths, and the central role of the menu state as a navigation hub. The dual-mode architecture allows for clear separation of concerns between security management and content editing functions.

8. Testing and Results

8.1 Synthesis Results

The Verilock design was synthesized using Vivado Design Suite targeting the Artix-7 XC7A35T FPGA. The synthesis completed successfully with no critical warnings. Resource utilization remained well within device limits, demonstrating the efficiency of the implementation.

Resource Utilization Summary:

- Look-Up Tables (LUTs): Approximately 2-3% of available resources
- Flip-Flops: Minimal utilization for state registers and counters
- Block RAM: Utilized for 512x5-bit message storage
- Timing: Positive slack achieved on all paths at 100 MHz

8.2 Functional Testing

Comprehensive functional testing was performed on the hardware to verify all system features:

Test Case 1: Default Password Authentication

The system was tested with the default password (six UP presses). Both SETP and WORD modes correctly authenticated with the default pattern and displayed appropriate success states.

Test Case 2: Password Change Operation

A new password pattern was successfully set through the SETP mode. The system correctly rejected the old password and accepted the new pattern for subsequent authentication attempts.

Test Case 3: Error Handling

Incorrect password entries correctly triggered the error state, displaying "ERRR" for one second before returning to the menu. The password entry counter displayed accurately during authentication.

Test Case 4: Message Editing

Character editing was verified for all 26 letters plus space and null characters. The cursor blink, viewport scrolling, and character cycling all functioned as designed.

Test Case 5: Navigation Modes

Both looping and infinite navigation modes were tested. Looping mode correctly skipped null characters, while infinite mode allowed access to all 512 positions. The SW0 LED indicator accurately reflected the current mode.

8.3 Display Verification

All 7-segment character patterns were verified for correct rendering. The multiplexed display showed no visible flicker, and the cursor blink rate was consistent and easily visible. Menu text ("SETP", "WORD", "AUTH", "DONE", "ERRR") displayed correctly in all states.

9. Future Enhancements

Several enhancements could extend the functionality and security of the Verilock system:

1. **Variable-Length Passwords:** Implementing support for passwords of varying length (4-8 directions) would increase security while allowing users to balance convenience with protection level.
2. **Scrolling Text Display:** Adding automatic scrolling functionality would allow viewing of longer messages without manual navigation, enhancing usability for message review.
3. **Extended Character Set:** Including numbers (0-9) and common punctuation marks would expand the range of messages that can be composed, making the system more practical for real-world applications.
4. **Multiple Message Storage:** Implementing separate memory banks for multiple messages with selection capability would transform the system into a more versatile note-keeping device.
5. **Attempt Limiting:** Adding a lockout mechanism after a specified number of failed authentication attempts would provide protection against brute-force attacks.
6. **RGB LED Feedback:** Utilizing the Basys3 RGB LEDs for status indication would provide more intuitive visual feedback, such as green for success, red for errors, and blue for editing mode.

10. Discussion and Conclusion

10.1 Discussion

The Verilock project successfully demonstrates the integration of multiple digital design concepts into a cohesive, functional system. The finite state machine architecture proved effective for managing the complex interactions between authentication, navigation, and editing functions. By maintaining a clear separation between states and well-defined transition conditions, the design achieved both reliability and maintainability.

The decision to implement edge detection without hardware debouncing simplified the design while remaining functional due to the Basys3 board's button conditioning. However, this approach may require revision for deployment on boards with raw button inputs. The password encoding scheme using 2-bit direction values represents an efficient use of storage while maintaining adequate security through pattern complexity.

The dual-mode navigation system addresses different user needs: looping mode provides efficient movement through populated message regions, while infinite mode enables precise positioning for sparse data entry. This flexibility demonstrates how configurable behavior can be implemented through simple hardware switches.

Resource utilization analysis reveals that the design operates well within the constraints of the Artix-7 FPGA, leaving substantial headroom for future enhancements. The positive timing slack indicates that the 100 MHz clock frequency could potentially be increased if higher performance were required.

10.2 Conclusion

The Verilock Password-Protected Word Display System successfully achieves its design objectives, demonstrating a practical implementation of secure message handling on an FPGA platform. The project showcases fundamental digital design principles including finite state machine design, memory management, display multiplexing, and user interface implementation through Verilog HDL.

Key accomplishments include a functional 6-direction password authentication system, a 512-character message buffer with efficient editing capabilities, intuitive 7-segment display rendering for 28 characters, and dual operational modes for flexible navigation. The modular architecture facilitates future enhancements and serves as a foundation for more complex secure display systems.

This project provided valuable hands-on experience in the complete FPGA development workflow, from specification through hardware validation. The skills developed in state machine design, timing analysis, and hardware debugging are directly applicable to more complex digital system projects. The Verilock system stands as both an educational exercise and a functional prototype demonstrating the capabilities of modern FPGA technology.

References

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Appendix A: Pin Assignments

The following table summarizes the FPGA pin assignments used in the Verilock constraints file:

Signal	Pin	Description
clk	W5	100 MHz system clock
BTNU	T18	Up button
BTND	U17	Down button
BTNL	W19	Left button
BTNR	T17	Right button
BTNC	U18	Center button
SW0	V17	Mode switch
seg[6:0]	W7-U7	7-segment cathodes
an[3:0]	U2, U4, V4, W4	7-segment anodes
dp	V7	Decimal point
led[15:0]	U16-L1	Status LEDs

Table 3: FPGA Pin Assignments